Reykjavík, Iceland | +354 848-1230
arnihafthor@gmail.com
Portfolio | GitHub | LinkedIn

Hafþór Árni Hermannsson

Mechatronics Engineer | Hardware Design & Development

Hands-on mechatronics engineer who enjoys designing, building and testing electromechanical systems. Experience includes PCB development, embedded programming, mechanical prototyping and field deployment of underwater camera and laser systems on fishing vessels, alongside a broad range of self-directed projects.

Experience

Optitog May 2024 – present
Mechatronics Engineer | Reykjavík, Iceland
Full-time initially; later part-time and contract work.

- Led electronics and embedded hardware development for underwater camera and laser modules deployed on working fishing trawls, including sensor and actuator integration, power regulation and thermal management.
- Main designer and maintainer of custom 4-layer PCBs in EasyEDA, responsible for subsequent board revisions, system integration, bring-up and debugging.
- Assembled, commissioned and troubleshot systems during deployments in Norway and Sweden, working in a two-person team to upgrade and repair hardware in the field.
- Built Python pipelines combining YOLO detections and sensor logs into reports that informed hardware revisions; co-developed an event-camera laser unit supported by the Icelandic Student Innovation Fund.

Media ehf Summer 2023
Mechatronics Engineer | Reykjavík, Iceland

- Developed a ceiling-mounted mmWave radar prototype for non-contact elderly monitoring, integrating fall detection and vital-sign monitoring with microcontrollers, radar modules, custom electronics and 3D-printed housings.

Selected Projects

Wave-Energy Harvester M.Sc. thesis · Publication 2026

- Independently designed, built and tested a wave-energy harvester to convert a monitoring buoy's motion into electrical power, covering mechanical design, electronics, software and experimental evaluation.
- Built a custom coil winder, motion and power instrumentation, and a large motorised wave simulator to reproduce buoy motion in ocean waves, with closed-loop position control and safety interlocks.
- Validated coupled electromagnetic and mechanical models against measurements and used them to evaluate scaled designs toward a 10 W buoy power target.

Invisible Drum Set GitHub 2024

- Developed a motion-controlled drum kit using two IMUs and an ATmega328P, with bare-metal C drivers, sensor fusion and drift recalibration; achieved 30–50 ms motion-to-sound latency.

NATO TIDE Hackathon GitHub 2026

- Built a working casualty-evacuation routing demo within the hackathon window, assigning casualties to hospitals by role of care and bed availability, with map-based route visualisation for operators.

Practical Work Experience

Summer and seasonal jobs alongside school since 2015, including **workshop assistant** at Vélfang, assembling and installing machinery; **deckhand** on a Norwegian longliner; **painter** at Nestak; **plumber's assistant** at Thor, including laying over 40 km of snow-melt piping beneath a football pitch; and three summers in fish processing at Síldarvinnslan.

Education

M.Sc. Mechatronics Engineering
Reykjavík University, 2023–2026
Graduated June 2026.

B.Sc. Mechatronics Engineering
Reykjavík University, 2020–2023

Skills

Electronics
Schematic capture and 4-layer PCB layout (EasyEDA, some KiCad), power regulation, motor and servo drive, sensor integration, soldering and rework, board bring-up and debugging.

Mechanical & fabrication
Fusion 360 (some Inventor), 3D printing, waterjet cutting, workshop fabrication, mechanism design and component sourcing.

Embedded & control
C/C++ (bare-metal and Arduino), ESP32, ATmega, STM32, nRF52840, Raspberry Pi and embedded Linux; I2C, SPI, UART; PID loops, state machines, safety interlocks and watchdogs.

Software & simulation
Python (OpenCV, pandas), MATLAB/Simulink, Linux/Bash, YOLO; Ansys Maxwell, FEM and physics modelling.

Self-Directed Builds

20+ documented home builds
Tesla coil with hand-wound secondary and custom driver boards, 3D-printed star-tracking camera mount, induction heater, high-voltage supply and IR theremin synthesizer. Portfolio

Raspberry Pi rover
Motor control, live video streaming, vision and LLM scene feedback. GitHub

nRF52840 2.4 GHz sniffer
Reverse-engineering proprietary wireless HID protocols. GitHub

bestavedrid.is
Live geospatial weather platform built with Flask and Leaflet.

Practical

Icelandic: native
English: fluent
Danish: basic

Driving licence: B

Available immediately.